```
long detRate(int recpin){
    long baud,   x;

    for (int i = 0; i < 5; i++){
        while(digitalRead(recpin) == 1){} // wait for low bit to start
        x = pulseIn(recpin,LOW);    // measure the next zero bit width
        rate = x < rate ? x : rate;
    }
     if (rate < 12)
        baud = 115200;
     else if (rate < 20)
        baud = 57600;
     else if (rate < 29)
        baud = 38400;
     else if (rate < 40)
        baud = 28800;
     else if (rate < 60)
        baud = 19200;
     else if (rate < 80)
        baud = 14400;
     else if (rate < 150)
        baud = 9600;
     else if (rate < 300)
        baud = 4800;
     else if (rate < 600)
        baud = 2400;
     else if (rate < 1200)
        baud = 1200;
     else if (rate < 2400)
        baud = 600;
     else if (rate < 4800)
        baud = 300;
     else
        baud = 0;
     return baud;
}
```

Fig. 2: A screenshot of the source code

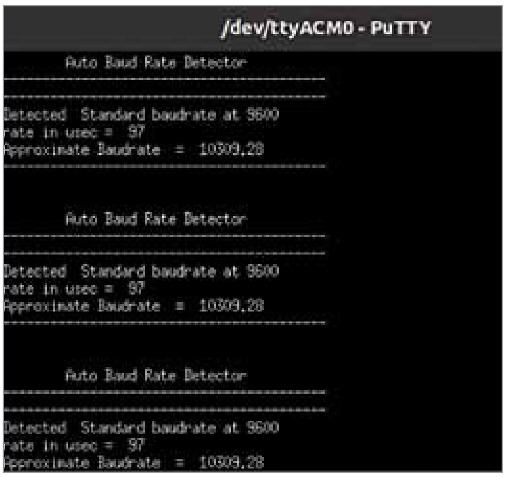

Fig. 3: Screenshot of serial console application (putty)

any serial device quickly. The following baud rates were detected and verified by the author with his prototype: 300, 600, 1200, 2400, 4800, 9600, 14400, 19200, 28800, 38400, 57600, 115200.

Please note, this baud rate detection algorithm may fail to predict if characters with multiple zero bits in a row are received. But that is a rare condition. The documentation for pulse in states that it can be used for intervals of 10 microseconds to 3 minutes.

The detector cannot be used to check baud rate above 115200. EFY

EFY Note
The source code of this project is available for free download at *www.electronicsforu.com*

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